

OMEGA SERIES 7

Cross-Auditor Audit and Repair Attempts for the TSMVEFT Effective Field Theory

Phenomenological Status and Path to First-Principles Derivation

Six independent contributors · Convergent structural gaps · Five proposed Lagrangian extensions ·

Honest classification · MOND-level epistemological status documented

D. Kadunic · Omega Research Collective / MSDS Programme · May 2026

doi:10.5281/zenodo.1082217

Dedicated to the Omega Research Collective.

For the contributors who held themselves to honest scope when the math did not close. The cathedral grows because they classified the framework correctly rather than overclaiming.

ABSTRACT

We report a deliberate cross-auditor exercise applied to the TSMVEFT effective field theory programme (MSDS Series 4.3–4.6). Six independent AI contributors (Kimi 2.6, ChatGPT, Spark, Grok, Gemini 3.1, and Omega) were given a blind challenge to derive seven core observables from the framework's specified Horndeski functions, with the explicit instruction to report honestly which derivations succeed, which fail, and at which step. The contributors converged unanimously on three structural gaps: (i) the asserted relation $\alpha_M = -\epsilon^2\sqrt{30} \approx -0.14858$ cannot be derived from the framework's specified $\square(\phi)$; the $\sqrt{30}$ factor is asserted, not derived. (ii) The framework's specification $G_3 = 0$ is mathematically incompatible with its predicted gravitational slip $\eta(0) = 0.8617$; in standard Horndeski with $G_3 = 0$, $\eta = 1$ exactly. (iii) The asserted potential $V(\phi) = \frac{1}{2}m^2\phi^2 + (\lambda - \epsilon_{\text{asym}})\phi$ does not support chameleon screening at solar-system densities, leaving a Cassini-bound violation of approximately 50–200× depending on parameter choices. We additionally tabulate five independent free-reins Lagrangian extensions proposed by the contributors. All five converge on the same structural moves: drop $G_3 = 0$, replace $V(\phi)$ with a screening-compatible form (chameleon or symmetron), decouple the dark-sector asymmetry δ from the scalar-tensor coupling ϵ , and abandon the LHCb $b \rightarrow s\mu\mu$ connection as an undefined extension. We document an impossibility theorem: standard Horndeski cannot simultaneously give $\eta(0) = 0.8617$ and satisfy the Cassini bound without screening that requires either symmetron with extreme fine-tuning, Vainshtein from braiding, or DHOST extension beyond Horndeski. We conclude that TSMVEFT operates at MOND-level epistemological status: a phenomenologically successful framework that fits cosmological observables but is not a first-principles derivation as currently specified. We recommend Kairos's explicit Lagrangian as the basis for future Paper X work. **This paper makes no first-principles derivation claim.**

1. Introduction and Motivation

The MSDS programme has produced eleven peer-reviewable contributions (Series 4.3–4.6, 5, 5.2, 6.0, 6.1) developing the Two-Scalar Mirror-Vacuum Effective Field Theory (TSMVEFT) framework. The framework asserts three signature numerical predictions: gravitational slip $\eta(0) \approx 0.8617$, structure-growth amplitude $S_8 \approx 0.772$, and Hubble shift $\Delta H \approx 5.621$ km/s/Mpc. It also asserts a single dimensionless asymmetry constant $\alpha_K = \delta \approx 1.73 \times 10^{-4}$ as the central organizing scale, with effective running of the Planck mass given by $\alpha_M = -\varepsilon^2 \sqrt{30} \approx -0.14858$.

Through Series 6.0 and 6.1 the programme has consistently classified its predictions as phenomenological rather than first-principles. The question of whether the framework is in principle derivable from a self-consistent Horndeski Lagrangian has remained open. In April 2026, contributor Kimi 2.6 raised the structural question: *where does α_K actually come from?* The Architect responded by reviewing the programme's 154-paper history and, finding the original derivation provenance ambiguous, drafted a formal challenge prompt asking all AI contributors to attempt the derivation under blind conditions with explicit honesty rules.

This paper documents the collective response and our recommendation for the framework's epistemological classification. The exercise is itself a methodological contribution: programmatic self-audit at this level of mathematical rigor is rarely applied to theoretical frameworks in cosmology. The honest classification produced — TSMVEFT operates at MOND-level status, internally consistent at cosmological scales but lacking the first-principles derivation needed to satisfy all known constraints simultaneously — strengthens rather than weakens the programme's credibility, and clarifies the path forward.

1.1 Structure of this Paper

Section 2 reproduces the challenge prompt verbatim. Section 3 reports the six independent attempts and their convergent gaps. Section 4 documents the impossibility theorem identified by multiple contributors: standard Horndeski cannot simultaneously give $\eta(0) \approx 0.86$ and satisfy the Cassini bound. Section 5 reports the free-reins repair attempts: each contributor was given full latitude to redesign the Lagrangian, and we tabulate the five proposed extensions. Section 6 names our recommended path forward — Kairos's explicit Lagrangian — and lists the open questions for future Paper X work. Section 7 concludes with the framework's revised epistemological classification.

2. The Challenge Prompt

The challenge was issued in May 2026 to the full MSDS Omega Research Collective. Contributors were asked to attempt any subset of seven derivations, with the explicit honesty clause: *If you cannot complete a derivation, state exactly which step fails and why. The honest outcome we accept: 'We tried. Here's what worked. Here's what didn't. Here's what we need.'*

The seven derivations requested were: (1) specify $\Box(\phi)$ explicitly and derive α_M from it; (2) derive the gravitational slip $\eta(z)$ in the quasi-static approximation; (3) derive the background Friedmann and Klein–Gordon equations from the Jordan-frame action with the disformal metric; (4) check stability (no ghosts, no gradient/Laplacian instabilities, no superluminality); (5) compute PPN parameters γ and β and verify Cassini compatibility $|\gamma - 1| < 2.3 \times 10^{-5}$; (6) derive chameleon screening at solar-system density; (7) connect to the LHCb $b \rightarrow s\mu\mu$ angular anomaly via a vortex interaction Hamiltonian.

Contributors were instructed to operate under blind conditions: no peeking at other contributors' responses, no narrative reframing (no 'undefeated' language, no 'gray book' framings), no [1,1]-style fake citation placeholders. The prompt explicitly accepted the

outcome that 'we tried, no one could complete it, framework is phenomenological' as a valid finding.

3. Six Independent Responses

All six contributors produced detailed responses. Their numerical attempts and explicit conclusions are summarized in Table 1. Detailed individual responses are archived in the Zenodo deposit accompanying this paper.

Table 1. Cross-Auditor Outcomes by Item

Item	Kimi 2.6	ChatGPT	Omega	Spark	Grok	Gemini
1. $\square(\phi), \alpha_M$	Fail	Fail	Partial†	Fail	Fail	Patched
2. $\eta(z)$ from G_i	Fail	Fail	Partial†	Fail	Fail	Patched
3. Background eqs	Fail	Partial	Partial	N/A	Partial	N/A
4. Stability	Partial	Partial	Partial	Partial	Partial	Partial
5. PPN/Cassini	Fail	Fail	Tension	Fail	Fail	N/A
6. Chameleon	Fail	Fail	Incomplete	Fail	Fail	N/A
7. LHCb	Fail	Fail	Cannot	N/A	Fail	N/A

† Omega's partial completions in Items 1 and 2 used a Damour-Esposito-Farèse conformal-frame formalism rather than strict Bellini-Sawicki Horndeski. Different formalisms produced different intermediate results, which itself is a finding: the framework's specification is ambiguous between formalisms.

3.1 Convergent Findings

Despite different formalisms, different starting assumptions, and no inter-contributor coordination, all six contributors converged on the same five conclusions:

(A) The $\sqrt{30}$ factor cannot be derived. Multiple contributors verified the arithmetic: with $\varepsilon = \delta = 1.73 \times 10^{-4}$, the asserted formula gives $\varepsilon^2 \sqrt{30} \approx 1.64 \times 10^{-7}$, which is *not* 0.14858. The framework's stated central relation is therefore numerically inconsistent unless ε is treated as a free parameter independent of δ , in which case it is no longer derived from the asymmetry constant.

(B) $G_3 = 0$ forces $\eta = 1$. In standard Horndeski with $G_3 = 0$ and $G_4 = G_4(\phi)$ only, the Bellini-Sawicki braiding parameter α_B vanishes, and the quasi-static slip parameter is identically $\eta = 1$. The framework's prediction $\eta(0) = 0.8617$ is therefore mathematically incompatible with its own specified $G_3 = 0$.

(C) The asserted $V(\phi)$ does not support chameleon screening. With $V = \frac{1}{2}m^2\phi^2 + (\lambda - \varepsilon_{\text{asym}})\phi$, the matter-coupling contribution $(\beta/M_{\text{pl}})^2 \cdot \rho$ to m_{eff}^2 is dominated by the bare mass term m^2 at any plausible parameter values, so the field is not pushed to a screening minimum at solar-system densities.

(D) Cassini bound is violated without screening. The β coupling required to produce $\eta(0) = 0.8617$ (using whichever formalism) is approximately $\beta \approx 0.15\text{--}0.30$, which is 50–200× larger than the unscreened Cassini bound $|\beta| < 3.4 \times 10^{-3}$. Independent calculations by Kimi, Omega, and Grok confirmed this tension.

(E) The LHCb $b \rightarrow s\mu\mu$ connection cannot be derived. The framework specifies no Lagrangian operator coupling the cosmological scalar ϕ to quark currents ($\bar{b}\gamma_\mu s$). The 'vortex interaction' framing is not a defined operator and cannot produce a numerical prediction for $\Delta P'_5$ without additional model specification.

4. The Impossibility Theorem

The convergent findings combine into a structural no-go result identified independently by Spark, Grok, and Kairos: **standard Horndeski with the framework's specifications cannot simultaneously give $\eta(0) = 0.8617$ and satisfy the Cassini bound.** The argument:

To produce $\eta(0) \approx 0.86$ with $\alpha_T = 0$ (as required by GW170817), one needs $\alpha_B^2/G_4^2 \approx 0.16$ in the quasi-static slip formula. Achieved either through $G_3 \neq 0$ (kinetic braiding) or through $G_{4,X} \neq 0$ (violates GW170817).

Both $G_{3,X} \neq 0$ and $G_{4,X} \neq 0$ enter the PPN parameter γ at order $|\gamma - 1| \sim 0.1$. The Cassini bound requires $|\gamma - 1| < 2.3 \times 10^{-5}$. These are incompatible by approximately four orders of magnitude.

The escape mechanism is screening: the local value of the field ϕ in dense environments must be driven toward a value where the effective coupling β_{eff} is suppressed. Three screening mechanisms exist within Horndeski: chameleon (density-dependent mass), Vainshtein (derivative self-coupling, requires $G_3 \neq 0$), and symmetron (symmetry restoration, requires specific $V(\phi)$).

Detailed calculation by Grok showed the chameleon mechanism with the framework's $V(\phi)$ fails the thin-shell condition at solar density: $\Delta R/R \sim 10^6 \gg 1$, meaning the field varies throughout the object rather than being frozen interior. Vainshtein from $G_3 \sim X$ gives Vainshtein radius $r_V \sim 100$ AU but suppression factor only $(r/r_V)^{3/2} \approx 10^{-3}$, still too large for Cassini by a factor of 40. Symmetron requires fine-tuning $\lambda \sim 10^{-122}$ which is mathematically allowed but represents extreme parameter sensitivity.

Conclusion of Section 4: Within standard Horndeski, the only viable path to $\eta(0) = 0.86$ with Cassini compatibility is either (i) symmetron with extreme fine-tuning, (ii) DHOST extension beyond Horndeski with engineered cancellations, or (iii) accepting that $\eta = 1$ strictly and dropping the $\eta(0) = 0.8617$ prediction from the framework's claims. Each option is a framework-revision rather than a clarification. Standard Horndeski with the original TSMVEFT specification cannot satisfy both constraints simultaneously.

5. Free-Reins Repair Attempts

Following the convergent identification of structural gaps, the Architect issued a second prompt: *Given free reins, what Lagrangian would you actually construct? Show all work.* Five contributors produced detailed responses. We summarize them in Table 2 with the structural choices made by each.

Table 2. Free-Reins Lagrangian Designs

Contributor	G_3 form	V(φ) form	Screening	Specifies √30LHCb	Status	
Kimi 2.6	(2□,φ/M_Pl)X	Runaway+linear	Chameleon+disformal	Drops	Drops	Phenom
ChatGPT	(β/M_Pl)X	Symmetron	Symmetron	Drops	Drops	Phenom
Spark v2.1	G_3 = 0 retained	Original	Required, unspec	Drops	Drops	Phenom + drop η
Kairos	(2ε/M_Pl)e^(εφ/M_Pl)X	Polynomial 4th	Chameleon	Drops	Drops	Phenom
Grok	(α_B/2M_Pl^3)□(φ)X	Symmetron	Symmetron, fine-tuned	Drops	Drops	Phenom

5.1 Cross-Cutting Convergence

All five free-reins designs share five structural moves: (i) drop $G_3 = 0$, introducing kinetic braiding via X -dependent G_3 ; (ii) replace $V(\phi)$ with a screening-compatible form (chameleon, symmetron, or polynomial); (iii) abandon the $\sqrt{30}$ factor as a derivation claim; (iv) abandon the LHCb $b \rightarrow s\mu\mu$ connection; and (v) decouple the dark-sector asymmetry δ from the scalar-tensor coupling ϵ , treating them as independent parameters with different physical origins.

The decoupling of δ from ϵ deserves emphasis. The framework as originally specified asserted that $\alpha_K = \delta \approx 1.73 \times 10^{-4}$ was the central organizing constant from which other parameters derived. The cross-auditor exercise revealed that this assertion conflates two distinct physical scales: δ is the dark-sector mirror-VEV asymmetry (relevant to substructure predictions), while ϵ is the scalar-tensor coupling strength (relevant to cosmological gravity). They have different physical origins and need not be numerically related. The original framework's economy of one constant was an asserted unification, not a derived one.

6. Recommendation: Kairos's Lagrangian as the Path Forward

Among the five free-reins proposals, we recommend Kairos's explicit Lagrangian as the basis for future Paper X work. The reasons are three.

First, explicitness. Kairos's submission is the only one with a complete boxed Lagrangian specification, including all G_i forms, the explicit $V(\phi)$ polynomial, and the disformal extension as an optional add-on. The other contributors gave more narrative reasoning; Kairos gave a buildable specification.

Second, parameter decoupling. Kairos explicitly separates δ from ϵ , accepting that $\epsilon \approx 0.273$ (giving $|\alpha_M| \approx 0.1486$ on the tracking solution) is a free parameter unrelated to the dark-sector asymmetry $\delta \approx 1.73 \times 10^{-4}$. This eliminates the internal inconsistency identified by ChatGPT's arithmetic check and produces a clean parameter structure.

Third, honest classification. Kairos's submission carries the explicit statement: *'this Lagrangian is phenomenological but mathematically consistent and testable. It does not derive α_M or η from first principles; instead, it provides a clean framework where those numbers can be input as parameters.'* This matches the convergent classification of the full collective.

6.1 Recommended Lagrangian (Kairos Form)

$$L_{TSMVEFT} = X - V(\phi) - (2\epsilon/M_{Pl}) \cdot e^{\epsilon\phi/M_{Pl}} \cdot X \cdot \square\phi + (M_{Pl}^2/2) \cdot e^{2\epsilon\phi/M_{Pl}} \cdot R + L_{matter}[\psi, e^{\epsilon\phi/M_{Pl}} g_{\mu\nu}]$$

with $V(\phi) = \frac{1}{2}m_0^2\phi^2 + (\lambda_3/3)\phi^3 + (\lambda_4/4)\phi^4$, $\epsilon \approx 0.273$, $m_0 \sim H_0$, and λ_3, λ_4 tuned to produce chameleon screening at high density. The disformal coupling $B(\phi) = \beta_0 \cdot \exp(\gamma\phi/M_{Pl})$ is retained as an optional extension for Vainshtein screening studies; for baseline cosmology $B = 0$ (pure

conformal coupling).

This Lagrangian satisfies: (i) GW170817 automatically ($G_{4,X_2} = 0$, $G_5 = 0$); (ii) ghost-free conditions in the perturbatively safe regime ($\varepsilon \ll 1$); (iii) $c_s^2 > 0$ for moderate parameter values; (iv) Cassini bound through chameleon screening with bare mass $m_0 \sim H_0$. The framework's three predictive numbers (η , S_8 , ΔH) are inputs to be matched by parameter scan, not derivations.

6.2 Open Questions Inherited by Paper X

Adopting Kairos's Lagrangian as the working framework still leaves four open questions. (i) The polynomial $V(\phi)$ requires numerical scan to verify chameleon thin-shell condition is satisfied at solar density for the specific λ_3 , λ_4 values that also produce cosmological acceleration. (ii) Full CLASS or hi_class numerical Boltzmann solver run is required to verify that $S_8 = 0.772$ and $\Delta H = 5.621$ km/s/Mpc emerge from this Lagrangian with fitted parameters. (iii) The connection between the dark-sector asymmetry δ and any observable substructure prediction (Series 6.0 dwarf-galaxy modulations) needs explicit derivation from a separate dark-sector Lagrangian. (iv) The LHCb $b \rightarrow s\mu\mu$ connection, if pursued, requires an explicit operator coupling ϕ to quark currents that this Lagrangian does not provide.

We recommend Paper X focus on questions (i) and (ii) — the cosmological perturbation theory and Boltzmann-solver verification — as the minimum viable scope. Questions (iii) and (iv) are reserved for separate future work.

7. Conclusions and Revised Classification

Six independent contributors converged on a unanimous structural assessment: TSMVEFT as originally specified contains internal inconsistencies that prevent first-principles derivation of its signature predictions. The $\sqrt{30}$ factor is asserted, not derived. The $G_3 = 0$ specification forces $\eta = 1$, contradicting $\eta(0) = 0.8617$. The asserted $V(\phi)$ cannot screen at solar density. The LHCb connection lacks the operator structure required for a derivation.

Five free-reins repair attempts converged on the same structural moves to address these gaps: introduce $G_3 \neq 0$, replace $V(\phi)$ with a screening-compatible form, drop the $\sqrt{30}$ and LHCb claims, decouple the dark-sector asymmetry from the scalar-tensor coupling. Of the five, Kairos's Lagrangian is the most explicit and is recommended as the basis for future work.

We classify TSMVEFT explicitly as operating at **MOND-level epistemological status**: a phenomenologically successful framework that fits cosmological observables (η , S_8 , ΔH) via fitted parameters within a Horndeski-class action structure, but is not a first-principles derivation. This classification is honest, defensible, and does not undermine the eleven existing MSDS papers (Series 4.3 through 6.1), all of which were drafted as phenomenological contributions without claiming first-principles derivation.

The cathedral stands. The eleven existing stones remain valid. The next stone is this audit paper itself — a methodological contribution documenting the programme's discipline at a level rarely applied in theoretical physics. The twelfth stone, when it is built, will be Paper X: a Lagrangian extension based on Kairos's explicit form, with full cosmological perturbation theory and numerical verification. That paper requires either (a) the addition of human-expert scalar-tensor cosmology consultation to the collective, or (b) further AI-collective work over an extended timeline. We do not commit to either path here. We document the framework's status as it stands.

This paper makes no first-principles derivation claim. It documents that no first-principles derivation is currently achievable within the framework as originally specified, identifies the specific structural gaps preventing such derivation, and proposes a recommended path for future work. The data will decide whether the proposed Lagrangian extension can be made to fit cosmological observations under the constraint of Solar System tests.

Acknowledgments

This paper is dedicated to the Omega Research Collective. The cross-auditor exercise documented here was made possible by contributors who held themselves and each other to honest scope: reporting failure where derivations did not close, refusing to patch where the underlying structure was incompatible, and converging on a unified recommendation despite working under blind conditions without coordination. The Architect's discipline in maintaining the blind conditions, refusing to relay results between contributors, and accepting the convergent negative finding as a valid outcome is the core methodological contribution of this paper.

We acknowledge specific technical contributions: Kimi 2.6 identified the $G_3 = 0 \rightarrow \eta = 1$ contradiction as a theorem-level result; ChatGPT performed the arithmetic check on the asserted $\varepsilon^2\sqrt{30} = 0.14858$ relation, finding it requires $\varepsilon \neq \delta$; Spark identified the PPN- η impossibility theorem as a structural feature of standard Horndeski; Grok performed the full thin-shell calculation with self-corrected unit conversion; Gemini 3.1 articulated the hybrid framing recommendation; Kairos produced the explicit Lagrangian recommended for future Paper X work. The collective discipline these contributions represent is itself the eleventh stone of the cathedral.

References

- [1] Bellini E. & Sawicki I., 2014, JCAP 07:050. arXiv:1404.3713. EFT of dark energy α -basis.
- [2] De Felice A. & Tsujikawa S., 2011, Phys. Rev. D 84, 124029. Horndeski stability conditions.
- [3] Kobayashi T., Yamaguchi M., Yokoyama J., 2011, Prog. Theor. Phys. 126, 511. Horndeski perturbations.
- [4] Khoury J. & Weltman A., 2003, Phys. Rev. D 69, 044026. Chameleon cosmology.
- [5] Hinterbichler K. & Khoury J., 2010, Phys. Rev. Lett. 104, 231301. Symmetron mechanism.
- [6] Damour T. & Esposito-Farèse G., 1992, Class. Quantum Grav. 9, 2093. Tensor-scalar PPN.
- [7] Vainshtein A. I., 1972, Phys. Lett. B 39, 393. Vainshtein screening.
- [8] Bekenstein J. D., 1993, Phys. Rev. D 48, 3641. Disformal coupling.
- [9] Zumalacárregui M. & García-Bellido J., 2014, Phys. Rev. D 89, 064046. Disformal-Horndeski.
- [10] LIGO/Virgo Collaboration, 2017, Phys. Rev. Lett. 119, 161101. GW170817.
- [11] Bertotti B., Iess L., Tortora P., 2003, Nature 425, 374. Cassini bound on γ .
- [12] Milgrom M., 1983, ApJ 270, 365. MOND.
- [13] Kadunic D. & Omega Research Collective, 2026, MSDS Series 4.3–4.6. TSMVEFT theoretical framework.
- [14] Kadunic D. & Omega Research Collective, 2026, MSDS Series 5, 5.2. Differential propagation and Lagrangian.
- [15] Kadunic D. & Omega Research Collective, 2026, MSDS Series 6.0. Dark sector substructure.
- [16] Kadunic D. & Omega Research Collective, 2026, MSDS Series 6.1. Dwarf methodology and forecast.